

Zehao Liu

Data Engineer • Machine Learning Engineer

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Professional Summary

Early-career engineer with a CS and mathematics foundation, focused on Data Engineer and Machine Learning Engineer work. I build reproducible pipelines and ML workflows through structured execution, fast learning, and product-aware delivery.

- Core work: ETL and live data systems alongside model training, evaluation, and deployment.
- Working pattern: learn quickly, structure the work, ship, and review with data and feedback.
- Recent delivery: MQTT-backed dashboards, packaged ML pipelines, and reproducible analysis workflows.

Professional Strengths

- Connect engineering work to business goals and user experience across the full delivery loop.
- Break complex tasks into structured plans with clear priorities and risk control.
- Use AI-assisted coding, research, and documentation with verification and explainability as guardrails.
- Support cross-functional teams through Jira, documentation, reporting, and client communication.
- Stay steady under pressure while balancing detail quality and response speed.

Education

The University of Sydney

Sydney, Australia

Master of Computer Science specialized in Data Science and AI

Feb 2025 - Aug 2026

- Data engineering, statistical modeling, machine learning, databases, and large-scale data analysis with reproducible Python workflows.

Wilfrid Laurier University

Waterloo, Canada

Honours Bachelor of Computer Science and Mathematics

Sep 2019 - Jun 2024

- Algorithms, statistics, databases, AI, and computational problem-solving with honours-level coursework.

Technical Focus

- Data engineering, SQL, databases, ETL, and reproducible pipelines.
- Machine learning, model evaluation, deployment, and reproducible ML workflows.
- Data analysis, dashboarding, reporting, and stakeholder-ready insight.
- AI-assisted development workflows with verification, documentation, and explainability guardrails.

Skills

Technologies: caret, CatBoost, Docker, ggplot2, GitHub Actions, JavaScript, Konva.js, LightGBM, Matplotlib, MQTT, Node.js, NumPy, Pandas, Python, R, Render, Scikit-learn, Streamlit, TensorFlow / Keras, Vercel, Vite, WebSocket, XGBoost

Projects

NEM Monitoring Dashboard | [GitHub](#) | [Demo](#)

Data Engineering

- Maintained a reproducible CSV ingestion and normalization pipeline with separated preparation and publishing stages for data integrity.
- Automated deployment and refresh workflows with Docker, GitHub Actions, and documented fallback behavior for collaborators.
- Performed data cleansing on facility and market inputs, preserving missing values explicitly and keeping each pipeline stage testable.

Infinite Canvas Studio | [GitHub](#) | [Demo](#)

Software Engineering

- I contributed across product scoping, frontend canvas implementation, Edit/Present mode workflows, WebSocket collaboration, documentation, testing, and deployment.
- The app supports temporary online rooms for collaboration, separate frontend and relay deployments, and export paths for JSON, single-file HTML, and PROJ folder formats.

Medical Image ML Algorithm | *GitHub*

AI / ML

- Performed large-scale processing of 32,000 training and 8,000 test histopathology tiles with normalization, held-out evaluation, and reproducible CLI workflows.
- Ensured data quality through confusion-matrix review, Grad-CAM interpretability checks, and structured error analysis across nine tissue classes.
- Maintained a documented Python package with cached GridSearch artifacts and command-line training entry points for repeatable analysis.

EduAttain Prediction | *GitHub*

Data Analytics

- Cleansed and verified 40,976 ABS microdata records by cross-checking metadata against 146 coded survey fields and resolving encoding mismatches.
- Handled missingness, class imbalance, and unknown survey codes before producing a cleaned modeling dataset for downstream analysis.
- Built reproducible R and Python benchmarking workflows with stratified evaluation and stakeholder-ready exploratory reports.